VIVA NOTE: VARIANCE-OPTIMAL HEDGING UNDER ROUGH VOLATILITY AND JUMPS

MAC F244 PROJECT TEAM
INTERNAL STUDY NOTE (NOT FOR SUBMISSION)

This note explains, in plain language, what we did and why (main reference: Shreve, *Stochastic Calculus for Finance II*, Ch. 4–6 and 11). Sections 2–4 are the mathematics to know; Section 5 describes the simulation model at the level needed; Section 6 gives our results; Section 8 prepares for questions. Every number is generated from our result files, so it matches the report.

1. The question in one paragraph.
You sell a call option and want to remove its risk by trading the underlying. In Black–Scholes this works perfectly if you rebalance continuously and for free. Real hedgers trade at most a few times a day, pay costs, and face markets that switch between calm and stress. We ask how often to rebalance, whether to trade more in stress, and how much a better hedge ratio helps. We test this in simulation, under a realistic model from Wang and Wang's paper using their code, and on NIFTY data with synthetic and traded options.

2. Black–Scholes delta hedging.
Under $\mathbb{Q}$ with zero rate, $dS = \sigma S\, dW$. For the call price $C(t,S)$, Itô's formula gives

$$dC = \big(C_t + \tfrac{1}{2}\sigma^2 S^2 C_{SS}\big)\, dt + C_S\, dS.$$

The Black–Scholes–Merton PDE says the bracket is zero, so $dC = C_S\, dS$: holding $\Delta = C_S$ shares replicates the call. Equivalently, by the martingale representation theorem, every $\mathbb{Q}$-martingale (here the option price) is a constant plus a stochastic integral against the Brownian motion, and hence against S. One source of risk and one traded asset: the market is complete.

3. What changes in practice (derivations to know).
(a) Discrete rebalancing. Hold Δ_k shares over an interval of length h, during which the stock returns $R_k = \Delta S/S$. The short call's P&L is $\Delta_k\, \Delta S - \Delta C$. Expand $\Delta C \approx C_t\, h + C_S\, \Delta S + \tfrac{1}{2}\Gamma\, \Delta S^2$, with $\Gamma = C_{SS}$, and use $C_t = -\tfrac{1}{2}\sigma^2 S^2 \Gamma$ from the PDE:

$$\varepsilon_k \approx \tfrac{1}{2}\Gamma_k S_k^2 \big(\sigma^2 h - R_k^2\big).$$

Since $\mathbb{E}[R_k^2] \approx \sigma^2 h$, the mean is about zero. Since $\mathrm{Var}(R_k^2) \approx 2\sigma^4 h^2$,

$$\mathrm{Var}(\varepsilon_k) \approx \tfrac{1}{2}\Gamma_k^2 S_k^4 \sigma^4 h^2.$$

Errors in different intervals are uncorrelated (martingale differences), so over the T/h intervals the extra variance is of order h: halving the interval roughly halves it. Our test suite checks this in a Black–Scholes simulation.
(b) Gamma P&L. If the delta is computed at an implied volatility σ_i but the stock moves with realised volatility σ_r, then $\mathbb{E}[R_k^2] \approx \sigma_r^2 h$, and the expected P&L per interval is

$$\tfrac{1}{2}\,\Gamma S^2 \big(\sigma_i^2 - \sigma_r^2\big)\, h.$$

The seller gains when implied volatility exceeds the volatility that follows. *Our check (test period 2019–2026):*

- India VIX was above the next month's realised volatility on 79% of days (mean 17.1% against 14.7%).
- Traded options' implied vol was above realised vol over their life for 73% of options (mean 15.7% against 14.5%).

Date: November 6, 2026.

- In variance terms (what the formula uses), implied exceeded realised by a factor 1.36 (VIX) and 1.20 (options) outside the COVID window. The crash reversed it (0.73 and 0.68).

(c) Costs. Each trade costs κ times the notional traded. Trading more often reduces the error in (a) but raises costs: a trade-off. With NIFTY's costs (up to 20 bp), daily rebalancing still wins.

4. Incomplete markets and the minimum-variance hedge.

Why the market becomes incomplete. With stochastic volatility (a second Brownian motion) or jumps of random size, there are more sources of risk than traded assets, so the martingale representation in terms of S alone fails and no stock position replicates the call.

Itô's formula with jumps (Shreve Ch. 11). For a process X with jumps $\Delta X_s = X_s - X_{s-}$,

$$f(X_t) = f(X_0) + \int_0^t f'(X_{s-})\,dX_s + \frac{1}{2}\int_0^t f''(X_{s-})\,d[X]^c_s + \sum_{0<s\le t}\big(f(X_s) - f(X_{s-}) - f'(X_{s-})\Delta X_s\big).$$

The sum replaces the linear (delta) approximation by the exact change of f at each jump. That gap is why a delta hedge cannot cover jumps: a jump moves the option and the stock by different relative amounts.

The minimum-variance hedge. Minimise $\mathbb{E}^{\mathbb{Q}}[(Y - x - \int_0^T \vartheta_t\,dS_t)^2]$ for the payoff $Y = (S_T - K)^+$. Let $M_t = \mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_t]$ (the option price) and decompose

$$Y = \mathbb{E}^{\mathbb{Q}}[Y] + \int_0^T \vartheta^*_t\,dS_t + L_T, \qquad \vartheta^*_t = \frac{d\langle M, S\rangle_t}{d\langle S\rangle_t},$$

where the residual L is uncorrelated with S (the Galtchouk–Kunita–Watanabe decomposition). Think of it as a regression: ϑ^* is the "beta" of option moves on stock moves, and L is what no stock position can explain. The three pieces (a constant, a mean-zero stochastic integral and a residual uncorrelated with S) are orthogonal, so by Pythagoras, for any (x, ϑ),

$$\mathbb{E}\Big[\Big(Y - x - \int \vartheta\,dS\Big)^2\Big] = (\mathbb{E}[Y] - x)^2 + \mathbb{E}\int_0^T (\vartheta^* - \vartheta)^2\,d\langle S\rangle + \mathbb{E}[L_T^2].$$

So the best initial capital is the price, the best holding is ϑ^*, and $\mathbb{E}[L_T^2]$ is the unhedgeable risk.

Example to derive on the board (Heston). Let $dS = \sqrt{v}\,S\,dW$ and $dv = \kappa_v(\theta - v)\,dt + \xi\sqrt{v}\,dB$ with $d\langle W, B\rangle = \rho\,dt$, and $M_t = C(t, S_t, v_t)$. Two-variable Itô gives $dM = C_S\,dS + C_v\,dv + (\dots)\,dt$, so $d\langle M, S\rangle = (C_S\,vS^2 + C_v\,\rho\xi vS)\,dt$ and $d\langle S\rangle = vS^2\,dt$:

$$\vartheta^* = C_S + \frac{\rho\,\xi\,C_v}{S}.$$

With $\rho < 0$ (volatility rises when prices fall) and $C_v > 0$, you hold fewer shares than the delta: a price fall comes with a volatility rise that partly protects the option's value. In jump models a similar covariance term appears for the jumps.

5. The simulation model.

The paper's rough Hawkes–Heston model (under $\mathbb{Q}$, zero rate) is

$$\frac{dS_t}{S_{t-}} = \sqrt{V_t}\,dW^S_t + (\text{compensated crash jumps}),$$

$$V_t = g_0(t) + \int_0^t K(t-s)\,dZ_s, \qquad K(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)},$$

where Z collects the variance's own shocks (correlated with the price through ρ) and the jumps.

- *Leverage.* $\rho < 0$: falling prices come with rising volatility.
- *Crash jumps.* A jump of size z multiplies the price by $e^{-\Lambda z} < 1$ and pushes the variance up.

- *Clustering.* Jumps arrive at a rate proportional to V, and each jump raises V, so jumps come in clusters.
- *Memory and roughness.* V depends on all past shocks with power-law weights K; with $\alpha < 1$ the paths are rough.
- *Compensated* jumps have their expected effect subtracted, so the price has no drift under $\mathbb{Q}$.

Parameter	Value	Meaning
α	0.527	roughness of the kernel $K(t) = t^{\alpha-1}/\Gamma(\alpha)$
ρ	-0.731	correlation between price and variance shocks (leverage)
b	-1.812	mean reversion of the variance (negative = pulls back)
c	0.115	vol-of-vol: size of Brownian shocks to the variance
Λ	0.276	a jump of mark z moves the log-price by $-\Lambda z$
β	0.049	slope of the initial curve $g_0(t) = V_0 + \beta t^\alpha/\Gamma(\alpha+1)$
V_0	0.0079	initial variance
κ, m	1, 1	jump marks: $\nu(dz) = \kappa e^{-z/m} dz/m$
S_0	1.0	initial price (prices are relative to S_0)

TABLE 1. Model parameters used in the paper and its code (initial volatility $\sqrt{V_0} \approx 8.9\%$).

How the hedge is computed. Wang and Wang show that ϑ^* has the covariance form of Section 4 (a Brownian part involving ρ, plus a jump part). They compute it with Fourier methods and a Riccati–Volterra equation. We use their published code and verified that it reproduces their results.

6. What we did and what we found.

(a) Replication. The official code gives a variance reduction against no hedge of 0.824 (GKW) and 0.726 (BS delta); our rerun gives 0.828 and 0.732. The difference is Monte Carlo noise. *Careful:* these two numbers come from the official code, not from the arXiv paper.

Hedge	Official output	Our rerun
Variance-optimal (GKW) hedge	0.824	0.828
Black–Scholes delta (implied vol)	0.726	0.732

TABLE 2. Replication: variance reduction vs. no hedge.

(b) Discrete rebalancing. If you trade only at $t_0 < \cdots < t_{N-1}$ and pay κ per unit of notional traded, the P&L of the hedged short call is

$$e = x + \sum_k \varphi_k(S_{k+1} - S_k) - \kappa \sum_k S_k|\varphi_k - \varphi_{k-1}| - \kappa S_N|\varphi_{N-1}| - Y.$$

Rebalancing dates	GKW hedge	BS delta
512	0.828 (0.012)	0.732 (0.026)
64	0.806 (0.013)	0.670 (0.031)
16	0.738 (0.016)	0.541 (0.033)
4	0.583 (0.021)	0.333 (0.036)
1	0.374 (0.019)	0.108 (0.036)

TABLE 3. Simulation: variance reduction (s.e.) against the number of rebalancing dates.

Findings.

- Error grows as trading becomes rarer, as Section 3(a) predicts.
- The GKW advantage over BS grows with rarer trading: the better hedge ratio matters more when the position is held longer.
- GKW beats BS at every cost level (z from -5.1 to -6.2).
- Smart rules (bands, variance-triggered) beat fixed schedules by 6%–31% at the same expected cost, but only when turnover is low. At the cost-optimal point the change is only -1.2% for GKW ($z = -0.5$), which is not significant.

(c) Data and regimes. The data are official NIFTY 50 closes and India VIX checked against NSE. Regime labels (calm/stress) are calibrated on 2010–2018 only and use data up to day t only. The main label is VIX above 20.16 (with hysteresis down to 17.88); the second is 21-day realised volatility above 17.4%. A statistical Markov-switching label is used only as a robustness check. Out of sample, stress days are followed by roughly twice the volatility (Table 4).

Variant	VIX (hysteresis)	21-day realised vol
Main (train to 2018)	1.94 [1.58, 2.44]	1.79 [1.37, 2.41]
Train to 2017	1.94 [1.57, 2.46]	1.80 [1.37, 2.38]
Train to 2019	2.01 [1.65, 2.54]	1.85 [1.40, 2.47]
Start 2008 (unverified VIX)	2.25 [1.52, 3.35]	1.86 [1.32, 2.64]

TABLE 4. Test period: future 5-day volatility after stress / after calm [95% CI].

(d) Backtests on NIFTY. We sell a one-month ATM call each day and delta-hedge it. Options are either synthetic (priced at VIX) or traded (NSE bhavcopy, hedged with futures).

- Daily rebalancing is best for costs up to 20 bp, and the regime rule picked in training is worse in testing.
- At equal turnover, regime rules win only because of the COVID crash.
- Inverted labels (trading less in stress) are clearly worse.
- The seller's average gain is the gamma P&L of Section 3(b).

	Picked on train vs daily		Equal turnover, 5 bp		
Variant	5 bp	20 bp	beat fixed	med. z	beat placebo
Own implied vol	+35.0% (2.1)	+15.1% (1.4)	48/48	−1.0	48/48
Own IV, excl. COVID window	+51.8% (2.0)	+18.8% (1.3)	17/48	0.4	41/48
India VIX delta	+33.7% (2.1)	+16.3% (1.5)	47/48	−1.0	46/48

TABLE 5. Traded-option backtest (test period).

7. How we made the results trustworthy.

- *Same paths for all rules* (common random numbers) and *paired* standard errors. Jumps make unpaired errors huge.
- *Equal-turnover comparison.* A rule that trades more always has lower error and higher cost, so we compare a rule with a coin-flip mixture of the two neighbouring fixed schedules that has the same expected turnover. The mixture is itself a valid strategy, so this is a fair benchmark.
- *No look-ahead.* Every decision at t uses data up to t. This is tested by truncating or scrambling future data and checking that past decisions do not change.
- *Train/test split.* Rules and thresholds are chosen on 2010–2018 and evaluated once on 2019–2026.
- *Placebos.* Inverted labels and year-old labels show whether any gain comes from the regime information.

- *Block bootstrap.* Options written on consecutive days overlap, so their errors are correlated. Resampling blocks of days keeps that dependence.
- *Criterion* $J = \mathrm{Var}(e) + (\mathbb{E}\,\mathit{cost})^2$ *on real data.* The seller's mean gain (about +18 bp per traded option) is gamma P&L, not hedging quality, so we remove it. J equals the simulation criterion when the premium is fair.
- *Data checks.* Expiring futures settle at the index close at all 194 expiries, put–call parity holds within ±28 bp, and the VIX data were reconciled against NSE.
- *Tests.* An automated test suite (`tests/`), including a hand-computed option and agreement with the official code to 15 digits.

8. Likely viva questions.

Q. *Why is Black–Scholes hedging perfect, and what breaks it here?*
A. Itô plus the PDE gives $dC = C_S\,dS$, so the delta replicates the option. In practice we trade discretely (Section 3a), pay costs, and face stochastic volatility and jumps, which make the market incomplete (Section 4).

Q. *Why does the error grow as rebalancing becomes rarer?*
A. Between trades the option moves nonlinearly (gamma) while the hedge is frozen. Each interval adds variance $\approx \frac{1}{2}\Gamma^2 S^4 \sigma^4 h^2$, so the total grows like h.

Q. *Why did the option seller make money on NIFTY?*
A. Gamma P&L: $\frac{1}{2}\Gamma S^2(\sigma_i^2 - \sigma_r^2)$. Implied volatility was above the volatility that followed most of the time (79% of days for VIX), except in the COVID crash.

Q. *Why is the market incomplete?*
A. There are more sources of risk (a variance shock and jumps of random size) than traded instruments (the stock).

Q. *Why is the optimal initial capital the option price?*
A. The gain $\int \vartheta\,dS$ has mean zero under $\mathbb{Q}$, so the error splits off $(\mathbb{E}[Y]-x)^2$, which is minimised at $x = \mathbb{E}^{\mathbb{Q}}[Y]$.

Q. *What is* $\langle M, S\rangle$*?*
A. The expected instantaneous covariance of the option price and the stock. The holding must be chosen before the move, so it uses this expected covariance, not the realised one.

Q. *Did you derive the paper's hedge formula?*
A. No. We used the authors' published implementation and checked that it reproduces their results. Our contribution is the analysis of discrete, costly and regime-aware hedging, and the NIFTY backtests.

Q. *Why not just compare RMSE across rules?*
A. Rules with more trading have both lower error and higher cost. Comparing at equal expected turnover (equal cost) separates timing skill from simply trading more.

Q. *Why the block bootstrap?*
A. Daily-written one-month options overlap by up to 20 days, so their errors are strongly correlated; i.i.d. standard errors would be far too small.

Q. *What does the COVID result mean?*
A. The apparent advantage of regime-aware rebalancing comes from one extreme episode. Excluding it, the advantage disappears. One crisis is too little evidence for a general effect.

Q. *Why not fit the paper's model to NIFTY?*
A. That means fitting the whole volatility smile with a specialised pricer: a project in itself. On real data we therefore use Black–Scholes and Black–76 deltas.

Q. *What would you do next?*
A. Make costs depend on the regime, compare the minimum-variance hedge of Section 4 with the delta on real options, and add static option hedges.

9. Glossary.

Delta, gamma: first and second derivative of the option price in the stock price. *Implied / realised volatility*: the volatility in the option's price / the volatility the stock actually showed. *Variance reduction*: $1 - \text{Var(hedged)}/\text{Var(unhedged)}$. *Turnover*: notional traded. *bp*: basis point, 0.01%. *Placebo*: the same rule fed meaningless information. *Look-ahead bias*: using data not yet available at decision time. *Common random numbers*: evaluating all strategies on identical simulated paths. *Bhavcopy*: NSE's daily end-of-day file of all derivatives contracts.